

Insurance Reserve Intelligence Platform

A Modular Physics- and Knowledge-Informed Framework for
Insurance Reserve Prediction, Evaluation and Optimization

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Abstract

Insurance reserve estimation is a fundamental actuarial task that supports pricing, solvency management, capital planning, and financial decision making. Traditional reserve calculations rely on solving actuarial differential equations for every policy, which can become computationally expensive when applied to large insurance portfolios or repeated scenario analyses.

This project presents the **Insurance Reserve Intelligence Platform**, a modular framework that combines classical actuarial methods with Physics-Informed Neural Networks (PINNs) and Knowledge-Informed Neural Networks (KINNs) for reserve prediction. The platform integrates synthetic policy generation, actuarial reserve calculation, model training, evaluation, sensitivity analysis, optimization, and visualization into a unified workflow. A configurable loss framework incorporating data, physics, boundary, monotonicity, and business-oriented constraints was developed to guide model learning while preserving actuarial consistency.

A comprehensive evaluation framework was designed to assess not only prediction accuracy but also physics compliance, boundary satisfaction, monotonicity, generalization, and reserve sensitivities. The project further extends the reserve predictor to support business applications including premium optimization, product design optimization, capital allocation, and portfolio analysis. The final model achieved an R^2 score of **0.9887**, with an MAE of **785.48** and an RMSE of **1212.76** on the test dataset, demonstrating strong agreement with the classical actuarial solver.

Overall, the project establishes a modular reserve intelligence platform that combines actuarial science with physics-aware machine learning, providing a scalable foundation for future insurance digital twin systems and intelligent actuarial decision-support applications.

1 Introduction

1.1 Why Insurance Reserve Intelligence Matters

Insurance reserves are one of the most important quantities in life insurance, influencing product pricing, solvency management, capital planning, regulatory compliance, and financial decision making. In practice, actuaries require more than a single reserve estimate. They often need to analyse how reserves change under different economic conditions, evaluate sensitivities to policy parameters, optimise product designs, and study portfolio-level behaviour. These requirements motivate the development of an *Insurance Reserve Intelligence Platform*, where reserve estimation serves as the foundation for a wider range of actuarial analytics.

Classical actuarial methods, particularly those based on Thiele’s differential equation, remain the industry standard for reserve valuation. Although these methods provide accurate and reliable reserve estimates, they require numerical solution of differential equations for every policy and every new scenario. This becomes computationally expensive for large insurance portfolios and repeated scenario analyses.

1.2 From Classical Reserve Solvers to Physics-Aware Learning

Recent advances in scientific machine learning have enabled the development of surrogate models that approximate complex numerical solvers while preserving their governing principles. Physics-Informed Neural Networks (PINNs) incorporate differential equations directly into the training process, while Knowledge-Informed Neural Networks (KINNs) further integrate domain knowledge and business constraints to improve learning.

In this project, these approaches are applied to actuarial reserve estimation by combining a classical actuarial solver with PINN and KINN models. Once trained, the neural network acts as a fast differentiable reserve predictor that can support reserve estimation as well as downstream actuarial analyses.

1.3 Project Scope

Rather than developing only a reserve prediction model, this work focuses on building a complete Insurance Reserve Intelligence Platform. The platform includes synthetic policy generation, actuarial reserve calculation, PINN and KINN training, model evaluation, sensitivity and elasticity analysis, visualization, and business-oriented optimization modules. Together, these components provide a modular framework that can support future insurance digital twin and actuarial decision-support applications.

1.4 Contributions

The major contributions of this project are summarized below.

1. Developed a modular Insurance Reserve Intelligence Platform integrating classical actuarial methods with PINN and KINN models for reserve prediction.
2. Designed a synthetic policy generation pipeline capable of producing realistic insurance portfolios using actuarial assumptions.
3. Developed a configurable training framework incorporating data, physics-informed, and knowledge-informed loss functions.
4. Built a comprehensive evaluation framework including validation metrics, sensitivity analysis, elasticity analysis, and model diagnostics.
5. Extended the trained reserve predictor to support premium optimization, product design optimization, capital allocation, and portfolio optimization, demonstrating the practical applicability of the framework.

2 Background

2.1 Insurance Reserves

Insurance reserves represent the estimated financial liability that an insurer must hold to meet future policyholder obligations. For life insurance products, reserves are accumulated throughout the policy term to ensure that future claims can be paid while maintaining the financial stability of the insurer.

Reserve estimation plays a critical role in several actuarial applications, including premium pricing, solvency assessment, capital management, financial reporting, and regulatory compliance. Since reserve values depend on multiple policy characteristics such as age, policy term, mortality assumptions, interest rates, premium, and sum assured, accurately modelling reserve behaviour is essential for effective risk management.

In practice, reserve calculations are performed repeatedly across large insurance portfolios and under different economic scenarios. Consequently, efficient reserve estimation methods are becoming increasingly important for modern insurance analytics.

2.2 Thiele’s Differential Equation

For continuous-time term life insurance, reserve evolution can be described using Thiele’s differential equation,

$$\frac{dV(t)}{dt} = rV(t) + P - \mu(S - V(t)), \quad (1)$$

where

- $V(t)$ is the reserve at time t ,
- r is the interest rate,
- P is the premium,
- μ is the mortality intensity,
- S is the sum assured.

The equation models the change in reserve over time by combining investment growth, premium inflow, and the expected liability arising from mortality. Since the reserve at policy maturity is known, the equation is solved backwards from the terminal condition

$$V(T) = 0.$$

In this project, the classical Thiele solver is used as the benchmark for generating reserve trajectories, which serve as the training targets for the neural network models.

2.3 Physics-Informed Neural Networks (PINNs)

Physics-Informed Neural Networks (PINNs) are neural networks trained using both observational data and the governing physical or mathematical equations of a system. Unlike conventional neural networks that rely solely on data, PINNs incorporate differential equation residuals directly into the loss function, encouraging the model to produce physically consistent predictions.

For insurance reserve estimation, the governing equation is Thiele’s differential equation. During training, the network minimizes both the prediction error and the PDE residual, allowing it to learn reserve trajectories while preserving actuarial consistency.

The PINN formulation is particularly attractive because it combines the flexibility of deep learning with the interpretability of classical actuarial models. Once trained, the model provides rapid reserve predictions without repeatedly solving the differential equation.

2.4 Knowledge-Informed Neural Networks (KINNs)

Knowledge-Informed Neural Networks (KINNs) extend the PINN framework by incorporating domain-specific knowledge in addition to physical constraints. Rather than relying solely on governing equations, KINNs also learn from expert knowledge, empirical observations, and business rules relevant to the application.

In actuarial reserve modelling, this knowledge includes expected monotonic relationships between reserve and policy variables, solvency considerations, reserve limits, and other insurance-specific constraints. These additional sources of information help guide the learning process,

particularly in regions where data may be sparse or where purely data-driven models may violate known actuarial behaviour.

The modular architecture developed in this project supports both PINN and KINN training, enabling the same reserve prediction framework to incorporate either physics-based constraints alone or additional knowledge-informed objectives.

3 Insurance Reserve Intelligence Platform

3.1 Platform Vision

The objective of the proposed platform is not simply to replace a classical actuarial reserve solver with a neural network. Instead, it aims to build an integrated reserve intelligence framework capable of supporting a wide range of actuarial and business workflows.

In practical insurance environments, reserve estimation is only one component of the overall decision-making process. Actuaries and risk managers frequently perform sensitivity studies, stress testing, product pricing, portfolio analysis, capital assessment, and scenario evaluation. These tasks often require repeated reserve calculations under changing assumptions, making computational efficiency an important consideration.

The proposed platform therefore combines classical actuarial methods with differentiable machine learning models to provide a unified framework that supports both reserve prediction and downstream analytical tasks.

3.2 Platform Architecture

The platform follows a modular architecture in which each component performs a specific role within the actuarial modelling pipeline. Classical actuarial methods are used to generate reliable reserve trajectories, while PINN and KINN models learn a differentiable approximation of the reserve function. The trained reserve predictor is subsequently used by evaluation, sensitivity analysis, optimization, and visualization modules.

This separation between data generation, model learning, evaluation, and business analytics makes the framework flexible and extensible. Individual modules can be improved or replaced without affecting the remaining components of the platform.

3.3 Core Functionalities

The current implementation provides several capabilities beyond reserve prediction.

- Generation of realistic synthetic insurance portfolios.
- Classical reserve valuation using Thiele’s differential equation.
- Physics-informed and knowledge-informed reserve prediction.
- Comprehensive validation using accuracy, physics consistency, boundary conditions, monotonicity, and generalization metrics.
- Sensitivity and elasticity analysis for actuarial interpretation.
- Optimization modules supporting premium pricing, product design, capital allocation, and portfolio analysis.
- Automated visualization and reporting for model interpretation.

3.4 Business Perspective

From a business standpoint, the platform transforms reserve estimation from an isolated numerical computation into a reusable analytical service. Once trained, the reserve predictor can rapidly evaluate large numbers of policies and scenarios without repeatedly solving the underlying differential equation.

This enables interactive actuarial workflows such as scenario analysis, reserve sensitivity studies, premium optimization, product comparison, and portfolio-level decision support. Although the current implementation focuses on term-life insurance, the modular design provides a foundation for extending the framework to additional insurance products and future insurance digital twin applications.

4 Framework Design

This chapter describes the design and implementation of the proposed Insurance Reserve Intelligence Platform. The framework consists of a sequence of interconnected modules that transform synthetic policy information into reserve predictions and subsequently support actuarial evaluation and optimization. Each component was designed independently to improve modularity, maintainability, and future extensibility.

4.1 Overall Framework

The proposed framework combines classical actuarial modelling with physics-aware machine learning to construct a differentiable reserve prediction system. Instead of replacing the actuarial reserve solver, the framework uses it to generate high-quality reserve trajectories that serve as supervisory information during neural network training.

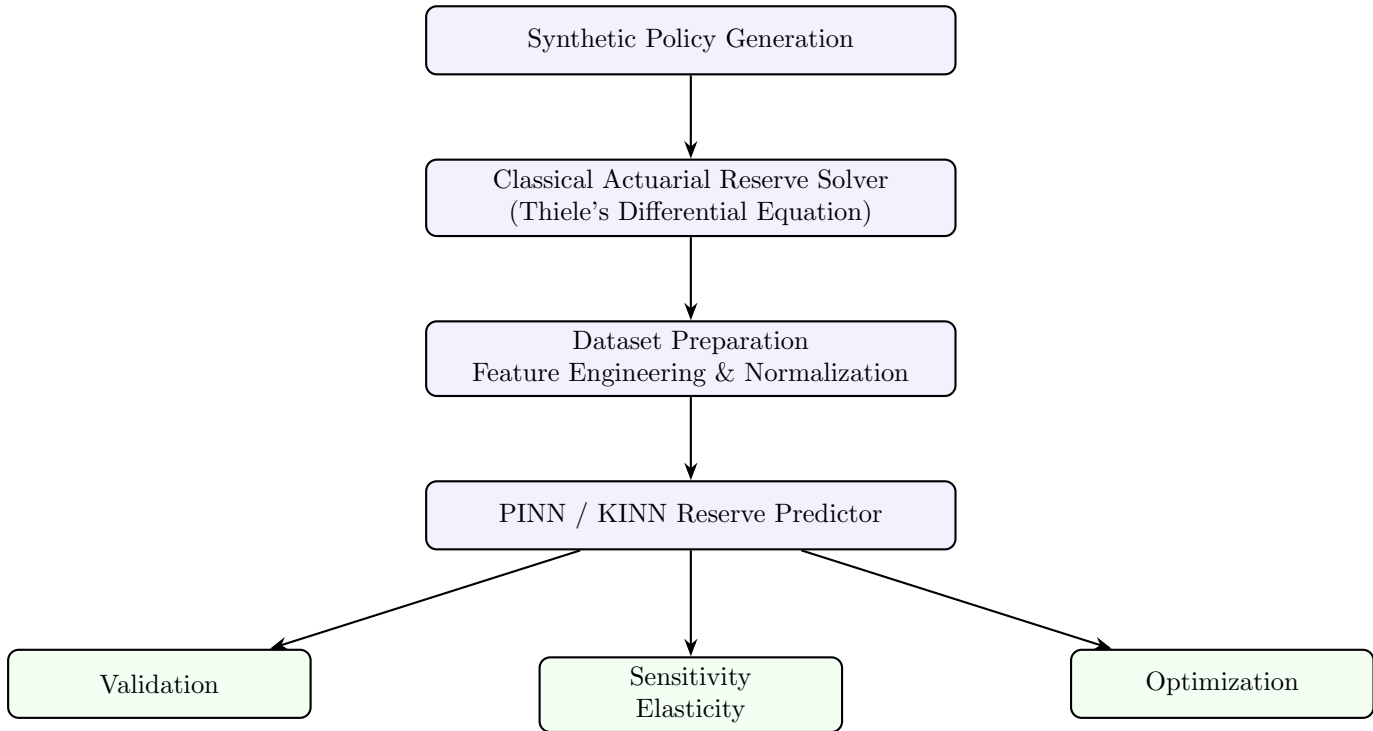


Figure 1: Overall architecture of the proposed Insurance Reserve Intelligence Platform.

The complete workflow begins with synthetic policy generation, followed by reserve computation using Thiele's differential equation. The resulting reserve trajectories are converted

into a supervised learning dataset, which is then used to train Physics-Informed and Knowledge-Informed Neural Networks. The trained reserve predictor is subsequently integrated with evaluation, sensitivity analysis, and optimization modules to provide an end-to-end reserve intelligence framework.

4.2 Synthetic Policy Generation

Obtaining real insurance policy data is often difficult due to confidentiality and regulatory restrictions. Therefore, a synthetic policy generation framework was developed to produce representative term-life insurance portfolios while preserving realistic actuarial relationships between policy variables.

Each generated policy consists of age, policy term, interest rate, mortality profile, premium, and sum assured. Rather than sampling these variables independently, actuarial assumptions are incorporated to maintain realistic policy characteristics. Premiums are computed using actuarial pricing principles, while mortality assumptions and policy constraints are applied to ensure consistency across the generated portfolio.

The synthetic dataset provides sufficient diversity for model training while remaining representative of practical insurance contracts.

4.3 Classical Actuarial Reserve Solver

Each synthetic policy is valued using a classical implementation of Thiele’s differential equation. The reserve trajectory is obtained by numerically integrating the governing reserve equation backwards from the terminal condition,

$$V(T) = 0.$$

The classical solver serves two important purposes. First, it generates the reserve trajectories used as supervisory targets during neural network training. Second, it acts as the actuarial benchmark against which reserve predictions and sensitivities are validated throughout the project.

Using the classical solver ensures that all learning is grounded in established actuarial principles rather than relying solely on observational data.

4.4 Dataset Preparation

The reserve trajectories produced by the actuarial solver are transformed into a supervised learning dataset by pairing policy features with reserve values at different time steps.

To improve numerical stability, all input features are normalized before training. The prediction target is represented as a standardized reserve ratio instead of raw reserve values, reducing scale differences between policies with different sum assured values. This representation enables stable optimization while preserving the underlying reserve dynamics.

The resulting dataset forms the common interface between the actuarial modelling components and the machine learning framework.

4.5 PINN and KINN Models

The reserve predictor is implemented as a fully connected neural network that receives policy features as input and estimates the corresponding reserve value.

Physics-Informed Neural Networks (PINNs) incorporate Thiele’s differential equation directly into the training objective, encouraging the predicted reserve trajectory to satisfy the governing actuarial dynamics. Knowledge-Informed Neural Networks (KINNs) further extend this framework by incorporating actuarial knowledge and business constraints during training.

The architecture is designed to support both approaches through a configurable loss framework, allowing different combinations of supervision to be explored without modifying the network itself.

4.6 Loss Framework

Rather than optimizing prediction accuracy alone, the reserve predictor is trained using a combination of complementary loss functions that enforce numerical accuracy, physical consistency, and actuarial knowledge.

The implemented losses are grouped into four categories.

- **Data Loss** supervises reserve prediction using the classical actuarial solver.
- **Physics-Informed Losses** enforce compliance with Thiele’s differential equation and terminal boundary conditions.
- **Knowledge-Informed Losses** encode actuarial expectations such as monotonic reserve behaviour, solvency constraints, reserve ceilings, and smooth reserve evolution.
- **Scenario-Oriented Losses** improve reserve behaviour under varying interest-rate environments by supervising scenario consistency and peak reserve behaviour.

This modular formulation enables additional business constraints to be incorporated into the training process with minimal changes to the overall framework.

4.7 Training Pipeline

Training begins by generating synthetic policies and computing their reserve trajectories using the classical actuarial solver. These reserve trajectories are transformed into normalized training samples before being supplied to the neural network.

During optimization, the model simultaneously minimizes supervised, physics-informed, knowledge-informed, and scenario-oriented losses. Model checkpoints are periodically saved and evaluated throughout training to monitor both prediction accuracy and actuarial behaviour.

This modular training pipeline enabled multiple experimental iterations during the project while preserving a consistent overall workflow.

The proposed framework combines synthetic policy generation, classical actuarial reserve modelling, PINN/KINN-based reserve prediction, and a configurable multi-objective training strategy within a unified architecture. This design provides a robust foundation for reserve estimation while supporting subsequent evaluation, sensitivity analysis, and optimization workflows presented in the following chapters.

5 Evaluation Framework

Accurate reserve prediction alone is insufficient for practical actuarial applications. A reserve model must also satisfy actuarial principles, preserve expected behaviour under changing policy assumptions, and provide reliable insights for downstream business tasks. To address these requirements, a comprehensive evaluation framework was developed to assess the proposed model from multiple perspectives.

Unlike conventional machine learning evaluation, which primarily focuses on regression accuracy, the proposed framework combines statistical performance metrics with physics-based validation, actuarial diagnostics, sensitivity analysis, and business-oriented evaluation. This enables a more complete understanding of model behaviour and helps identify limitations that may not be apparent through prediction error alone.

5.1 Prediction Accuracy

The first stage of evaluation measures how closely the predicted reserve values match the reserve trajectories generated by the classical actuarial solver.

The following regression metrics are computed on the test dataset.

- Mean Squared Error (MSE)
- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Coefficient of Determination (R^2)

These metrics provide an overall measure of numerical agreement between the learned reserve function and the actuarial benchmark.

Table 1: Final Prediction Performance

Metric	Value
Mean Squared Error (MSE)	1,470,771.88
Mean Absolute Error (MAE)	785.48
Root Mean Squared Error (RMSE)	1212.76
Coefficient of Determination (R^2)	0.9887

5.2 Validation Framework

Prediction accuracy does not necessarily imply that the learned reserve function satisfies actuarial principles. Therefore, additional validation diagnostics were developed to evaluate different aspects of model behaviour.

The validation framework assesses reserve prediction accuracy, compliance with Thiele’s differential equation, terminal boundary behaviour, monotonic reserve behaviour, and model generalization. These complementary diagnostics provide a more comprehensive assessment of reserve quality than conventional regression metrics alone.

5.3 Sensitivity Analysis

Sensitivity analysis evaluates how reserve values change when individual policy parameters are perturbed.

A reference policy is selected, and one feature is varied at a time while all remaining policy characteristics are held constant. This approach computes true partial derivatives of the reserve with respect to interest rate, mortality, premium ratio, and sum assured.

To verify the learned reserve behaviour, the sensitivities produced by the PINN/KINN model are directly compared with finite-difference sensitivities obtained from the classical actuarial solver.

5.4 Elasticity Analysis

While sensitivity values quantify the absolute effect of each policy parameter on reserve, they are expressed in different physical units and therefore cannot be compared directly.

Elasticity analysis normalizes each sensitivity into a dimensionless quantity, enabling comparison of the relative importance of different policy variables. This provides a more interpretable measure of reserve behaviour across parameters with different numerical scales.

5.5 Optimization Framework

One of the primary advantages of a differentiable reserve predictor is that it can be directly integrated into optimization workflows. Instead of repeatedly solving Thiele’s differential equation for every candidate solution, optimization algorithms query the trained reserve model to rapidly evaluate reserve values during the optimization process.

Several optimization modules were developed to demonstrate the practical applicability of the proposed framework. These include premium optimization, product design optimization, capital allocation, and portfolio optimization. Each module is formulated as a constrained optimization problem in which reserve values are predicted by the trained PINN/KINN model while satisfying relevant actuarial and business constraints such as premium bounds, reserve positivity, and solvency requirements. By replacing repeated numerical reserve calculations with neural network inference, the framework significantly improves computational efficiency while enabling interactive actuarial decision-making.

5.6 Business-Oriented Validation

Beyond numerical accuracy, the reserve predictor was evaluated from a business perspective to ensure that it behaved consistently under realistic actuarial scenarios.

Particular attention was given to reserve responses under varying interest-rate environments. Additional scenario-oriented loss functions, including the Interest Rate Scenario Loss and Interest Rate Peak Loss, were incorporated to improve the model’s ability to capture reserve behaviour across different economic conditions.

These experiments demonstrate that business-oriented validation can provide valuable insights beyond traditional machine learning performance metrics.

Summary

The proposed evaluation framework combines statistical accuracy, actuarial diagnostics, sensitivity analysis, elasticity analysis, and business-oriented validation within a unified workflow. Together, these evaluations provide confidence that the trained reserve predictor is suitable not only for reserve estimation but also for downstream analytical and optimization tasks.

6 Results and Discussion

This chapter presents the performance of the proposed Insurance Reserve Intelligence Platform and discusses the effectiveness of the developed framework. The evaluation focuses on reserve prediction accuracy, actuarial validation, sensitivity behaviour, and business applicability. All experiments were performed using synthetically generated term-life insurance policies, with reserve trajectories computed using the classical Thiele solver serving as the reference benchmark.

6.1 Reserve Prediction Performance

The proposed PINN/KINN framework was trained to approximate reserve trajectories generated by the classical actuarial solver. Prediction accuracy was evaluated on an independent test dataset using standard regression metrics.

The model achieved an R^2 score of **0.9887**, indicating excellent agreement with the classical reserve solver. The corresponding Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Squared Error (MSE) are presented in Table 1. These results demonstrate that the proposed framework accurately reproduces reserve trajectories across a diverse range of synthetic insurance policies.

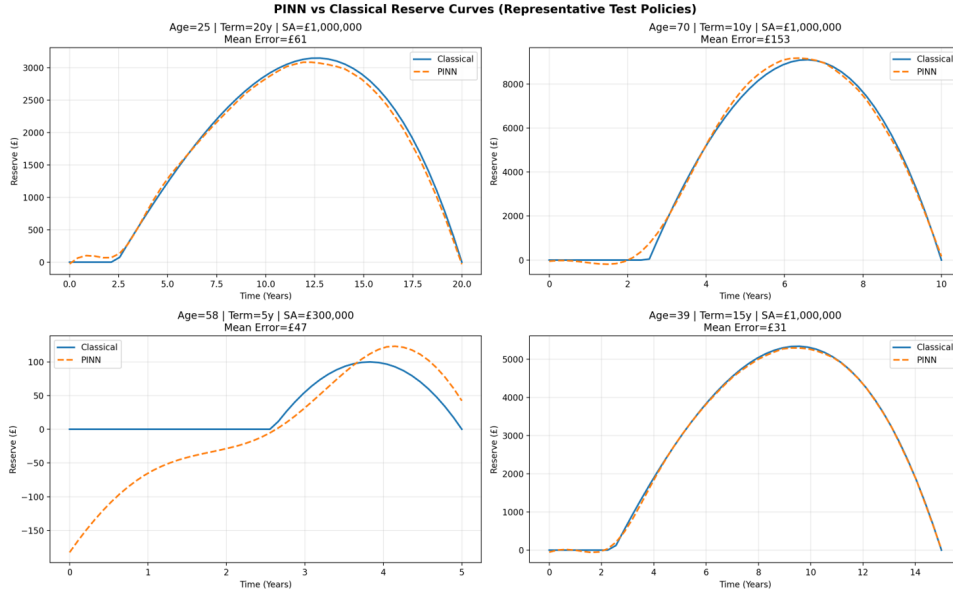


Figure 2: Comparison between reserve trajectories predicted by the proposed PINN/KINN framework and the classical actuarial solver.

6.2 Validation Results

Prediction accuracy alone does not guarantee that the learned reserve function satisfies actuarial principles. Therefore, additional validation diagnostics were used to evaluate physics consistency, boundary condition satisfaction, monotonic reserve behaviour, and generalization.

The validation report shown in Figure 3 summarizes these complementary evaluation metrics. Together, these diagnostics indicate that the proposed framework not only predicts reserve values accurately but also preserves the actuarial characteristics expected from the underlying reserve model.

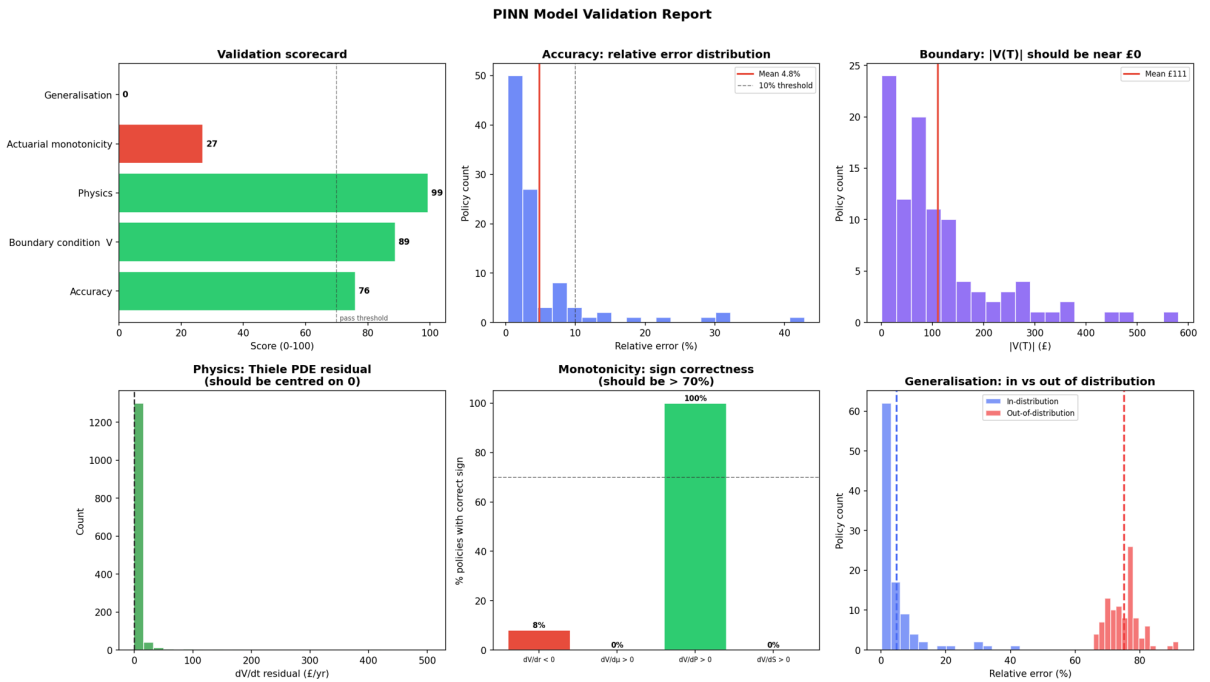


Figure 3: Validation report summarizing prediction accuracy, physics consistency, boundary behaviour, monotonicity, and generalization performance.

6.3 Sensitivity and Elasticity Analysis

Sensitivity analysis was performed to evaluate how reserve values respond to changes in individual policy parameters. A finite-difference approach was adopted in which each feature was perturbed independently while keeping all remaining variables fixed. This enables computation of true partial derivatives and facilitates direct comparison with sensitivities obtained from the classical actuarial solver.

To improve interpretability, elasticity analysis was also performed by normalizing sensitivities into dimensionless quantities. Together, these analyses provide valuable insight into the behaviour of the learned reserve function beyond prediction accuracy.

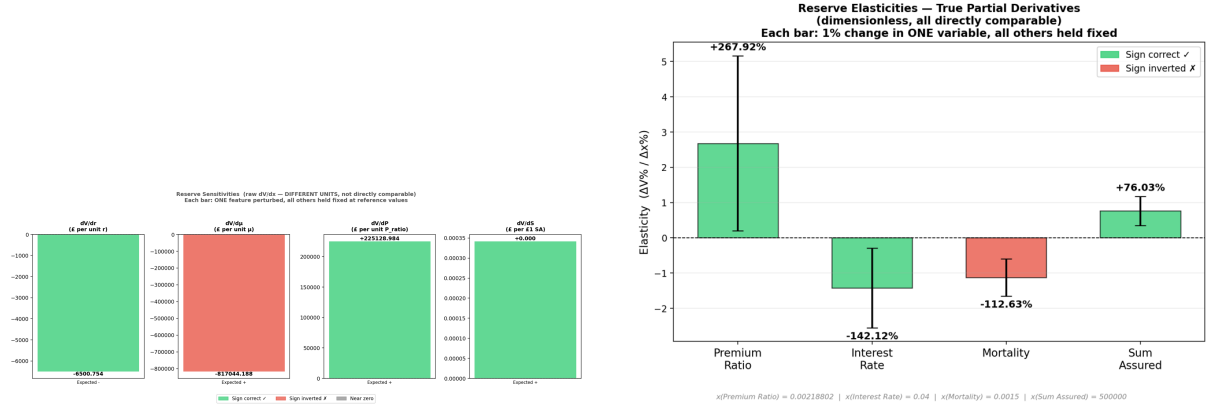


Figure 4: Sensitivity and elasticity analysis of the trained reserve predictor.

6.4 Optimization Applications

Beyond reserve prediction, the proposed framework demonstrates the practical value of differentiable reserve modelling through several optimization applications.

Premium optimization determines premium values that satisfy specified reserve objectives while respecting actuarial constraints. Product design optimization simultaneously adjusts multiple policy parameters to identify suitable product configurations. Capital allocation supports the efficient distribution of reserves across policies, while portfolio optimization evaluates reserve behaviour over collections of policies rather than individual contracts.

These applications illustrate how the trained reserve predictor can replace repeated numerical valuation during optimization, enabling significantly faster exploration of actuarial decision spaces. The results demonstrate that the proposed framework serves not only as a reserve prediction model but also as a computational engine for practical insurance analytics.

6.5 Discussion

The results demonstrate that the proposed framework successfully combines classical actuarial modelling with physics-informed and knowledge-informed learning. High prediction accuracy, together with comprehensive validation and sensitivity analysis, indicates that the learned reserve function captures both numerical and actuarial characteristics of the underlying reserve model.

The modular architecture further enables the reserve predictor to support downstream analytical tasks, including sensitivity analysis and optimization, without requiring repeated numerical solution of the governing differential equation. This significantly broadens the applicability of the framework compared with conventional reserve prediction models.

7 Limitations

Although the proposed Insurance Reserve Intelligence Platform demonstrates strong reserve prediction performance and supports several downstream actuarial applications, certain limitations remain.

- **Synthetic Dataset:** The framework has been developed and evaluated using synthetically generated term-life insurance policies. While actuarial assumptions were incorporated to generate realistic policy characteristics, validation using real-world insurance datasets is necessary to assess industrial applicability.
- **Limited Product Scope:** The current implementation focuses exclusively on term-life insurance. Extending the framework to more complex products, such as whole-life, endowment, annuity, and health insurance contracts, will require modifications to both the reserve equations and data generation process.
- **Scenario Coverage:** Although interest-rate scenario behaviour was improved through dedicated loss functions, the framework currently considers a limited set of economic and actuarial scenarios. More comprehensive stress testing involving inflation, lapse rates, policyholder behaviour, and correlated market variables remains future work.
- **Model Interpretability:** While sensitivity and elasticity analyses provide useful insights into reserve behaviour, further work is required to improve model explainability through uncertainty quantification, feature attribution techniques, and confidence estimation.
- **Optimization Validation:** The optimization modules demonstrate the practical applicability of the differentiable reserve predictor; however, large-scale validation using realistic insurance portfolios and operational business constraints has not yet been performed.

Despite these limitations, the proposed framework establishes a modular foundation that integrates actuarial modelling, physics-informed learning, evaluation, and optimization within a unified reserve intelligence platform.

8 Future Work

The current implementation establishes the foundation of an Insurance Reserve Intelligence Platform; however, several opportunities exist for further development.

8.1 Advanced Risk Analytics

Future work may extend the current sensitivity framework through second-order sensitivity analysis, interaction analysis between multiple policy variables, uncertainty quantification, and confidence interval estimation. These capabilities would provide deeper insight into reserve behaviour under changing actuarial assumptions.

8.2 Business Scenario Analysis

The framework can be expanded to support large-scale scenario analysis, including stress testing under adverse economic conditions, multi-factor what-if analysis, and reserve forecasting under changing mortality and interest-rate environments. Such capabilities would enhance the platform’s usefulness for actuarial risk management and regulatory reporting.

8.3 Enterprise-Level Decision Support

Future extensions may include portfolio-level reserve estimation, reserve movement attribution, production model monitoring, and model drift detection. These additions would enable the framework to support enterprise-scale insurance operations.

8.4 Insurance Digital Twin

The long-term vision of this work is the development of a comprehensive insurance digital twin capable of continuously monitoring insurance portfolios, forecasting reserve evolution, performing real-time scenario analysis, and supporting AI-assisted actuarial decision-making under dynamic economic conditions.

9 Conclusion

This paper presented the design and implementation of an Insurance Reserve Intelligence Platform that integrates classical actuarial reserve modelling with Physics-Informed and Knowledge-Informed Neural Networks.

A modular framework was developed covering synthetic policy generation, actuarial reserve computation, reserve prediction, model evaluation, sensitivity analysis, optimization, and visualization. The proposed framework achieved strong agreement with the classical actuarial solver while maintaining actuarial consistency through physics-informed and knowledge-informed learning.

Beyond reserve prediction, the developed platform demonstrates the potential of differentiable reserve models for practical actuarial analytics, including sensitivity analysis, optimization, and decision support. The proposed framework therefore provides a scalable foundation for future research and industrial applications involving insurance digital twins and intelligent reserve management systems.

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